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Finite slice analysis (FINA)—A general reconstruction method for velocity mapped and time-sliced ion imaging

J. O. F. Thompson, C. Amarasinghe, C. D. Foley, and A. G. Suits^{a)}

Department of Chemistry, University of Missouri, Columbia, Missouri, 65211, USA

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Since the advent of ion imaging, one of the key issues in the field has been creating methods to reconstruct the initial 3D distribution of particles from its 2D projection. This has led to the development of a number of different numerical methods and fitting techniques to solve this fundamental issue in imaging. In recent years, slice-imaging methods have been developed that permit direct recording of the 3D distribution, i.e., a thin slice of the recoiling fragment distribution. However, in practice, most slice imaging experiments achieve a velocity slice width of around 10%–25% around the center of the distribution. This still carries significant out-of-plane elements that can blur the spectrum, lose fine resolution, and underestimate the contribution from slow recoiling products. To overcome these limitations, we developed a new numerical method to remove these out-of-plane elements from a sliced image. The finite sliced analysis method models the off-axis elements of the 3D particle distribution through the use of radial basis functions. Once applied, the method reconstructs the underlying central slice of the 3D particle distribution. The approach may be applied to arbitrarily sliced or unsliced data and has the further advantage that it neither requires nor enforces full cylindrical symmetry of the data. We demonstrate this reconstruction approach with a broad range of synthetic and experimental data that, at the same time, allows us to examine the impact of finite slicing on the recovered distributions in detail. *Published by AIP Publishing.* [<http://dx.doi.org/10.1063/1.4979305>]

I. INTRODUCTION

Over the last three decades, studies of photodissociation and molecular collision dynamics have been dominated by the use of product imaging techniques. Ion imaging was first demonstrated by Chandler and Houston¹ and later improved by Eppink and Parker² with the high-resolution Velocity Map Imaging (VMI) technique. This, with some variation, has become a standard in studying a wide range of chemical phenomena and has been adapted to photoelectron imaging as well. However, these techniques suffer from one drawback, which is the necessity for the reconstruction of the 3D distribution from its 2D projection onto the imaging detector. As such, energy and angular resolved information contained within the central slice of this 3D Newton sphere is blurred by out-of-plane elements. Therefore, to gain meaningful insight from this technique, and produce high quality spectra, reconstruction of the 3D distribution is required. These reconstruction methods have some disadvantages, however, such as the introduction of noise along the centerline and a requirement for cylindrical symmetry in the data. To overcome these shortcomings and measure the central slice of the distribution directly, slice-imaging methods were developed in the early 2000s.^{3,4} Although these methods provide extraordinary clarity and approach a direct measurement of the 3D distribution, they nevertheless include some out-of-plane contributions that may blur or distort the inferred 3D distributions. We here describe a

new approach to the analysis of these images that also permits the reconstruction of unsliced images without requiring full cylindrical symmetry. At the same time, this method allows us to perform a detailed investigation of the impact of slicing on the inferred distributions and to compare the performance of slice imaging with conventional imaging under optimal conditions.

Since the advent of ion imaging and VMI, there have been numerous methods developed to reconstruct the 3D distribution from projected ion images. The initial approach employed the inverse Abel transform,^{5,6} in which an integral transform of the image is performed to obtain the central slice of the original 3D distribution. For this approach, the most visually striking disadvantage is the noise that accumulates along the centerline, the assumed cylindrical symmetry axis. This causes a loss of experimental resolution and reduces the accuracy of the angular distribution in the image, which is one of the key applications of photofragment imaging. Second, the transform assumes that the image is cylindrically symmetric about the central vertical axis of the image. This limits the potential experimental geometries for which this approach is mathematically valid. In fact, to our knowledge all approaches to-date assume cylindrical symmetry of the underlying distribution. However, with some simple parameter optimization, this approach can be used as an on-the-fly method for image inversion where speed, rather than resolution, is the most important criterion for choosing a method.^{7,8} Another early approach used was the Vrakking iterative method,⁹ which placed the noise of the transformation into the region of low velocity as opposed to the axis of cylindrical symmetry. However, this procedure requires intimate knowledge of the angular

^{a)} Author to whom correspondence should be addressed. Electronic mail: suitsa@missouri.edu

distribution for the technique to be effective and is very computationally demanding. In addition to these two methods, there was another early approach of note termed as “Onion-peeling.”¹⁰ In this method, the out-of-plane contribution of the original 3D particle distribution is removed by the use of a basis function which models and effectively removes the contribution inwards and effectively “peels away” layers of contribution until the central slice is all that remains. Recent work by Rallis *et al.*¹¹ has significantly improved the processing time of the onion peeling method, down to 50 ms for a small image and 0.6 s for a 1-megapixel image. Finally, a method was devised by Reisler and co-workers¹² to reconstruct the image using a combination of the Abel transform with a Gaussian basis-set expansion, commonly known as BASEX. In this approach, a set of forward basis functions with analytical inverse Abel transforms is fitted over the image to reconstruct it. This method handles noisy data better than the simple inverse Abel or onion-peeling methods and the reconstructed data are typically of higher resolution than the previously described methods. It has since become the most widely used approach. However, the BASEX method still naturally symmetrizes the 3D distribution and noise appears, growing toward the centerline where the data are less well characterized. More recently, a new and novel approach to image reconstruction was performed using the maximum entropy concept, where the most likely distribution to generate the recorded distribution is generated.¹³

Two variations on the methods above are polar onion peeling (POP)¹⁴ and polar BASEX (p-BASEX)¹⁵ approaches. Both of these techniques operate in polar coordinates, which better relate to the symmetry of the matter-light interaction in the gas-phase rather than the Cartesian grid related to the camera used in image acquisition. Working in polar co-ordinates has a number of advantages. First, conversion to polar co-ordinates is ultimately necessary to convert the image into a spectrum, so performing this transformation and the resulting interpolation prior to reconstruction, where the signal to noise level is significantly better, improves the resolution and signal-to-noise (S/N) of the resulting integrated spectrum. Second, the distributions are inherently described in a spherical framework, and the slicing operation is readily described in spherical coordinates. Finally, the transformation allows for a simple direct measurement of the product angular distribution prior to reconstruction, which is needed for aspects of the approach in which some preliminary information about the distribution is required to fit the data.

In 2001, Kitsopoulos and co-workers^{4,16} introduced slice imaging, in which the ion cloud is allowed to expand for a fraction of a microsecond after which it is accelerated to the detector. This expansion leads to an extensive dilation of the ion packet arrival time at the detector so that the central slice can be gated for recording directly. They noted numerous advantages of the approach, including the elimination of centerline noise in the reconstructed distributions, improved S/N, application to non-cylindrically symmetric geometries, and the possibility of imaging expanded regions of the image without the need for knowledge of the faster fragments. However, this approach required using mesh grids to keep the fields uniform in the initial acceleration and sacrificed much of the

advantage of VMI. In addition, the field-free expansion period is prone to space-charge distortion if many ions are present, so the signal levels must be reduced to prevent this. In 2003 Townsend *et al.* developed an alternative approach to slice imaging simply using low acceleration fields in the interaction region and an additional lens or two in the VMI lens array to accelerate and refocus the ion cloud.³ This approach, termed “direct current” or DC sliced imaging, had the further advantage that it gave an order of magnitude improvement in velocity focusing compared to basic VMI, rather than undermining it. Velocity resolution better than 0.5% has been reported using this approach.¹⁷ An analogous approach was reported at the same time by Liu and co-workers, which focused mainly on scattering experiments.¹⁸

Slice imaging has been adapted for a number of different experiments including the use of slice imaging for crossed-beam scattering,^{18–23} specific machines designed for H-atom photodissociation,²⁴ dissociative electron detachment studies,²⁵ Rydberg tagging,²⁶ and reflectron VMI²⁷ to name a few. Some of these involve recording both the position of the event using a 2D imaging detector and also the time of the event to allow a full 3D description of the particle momentum distribution to be recorded. An early example of such an approach is the application of wedge-and-strip or delay line anodes.^{28–31} These are frequently used in coincidence experiments with low signal counts but high repetition rates. They are capable of generating a full 3D distribution of events^{32,33} and allow for narrow slicing of the image, with precision down to 100 ps, but cannot readily tolerate the peak count rates seen in imaging experiments at low repetition rates. In recent years, new “intelligent” detector technologies have been developed that solve some of the issues with the delay line anode detectors by effectively moving the technique to individual channels within the detector.^{34–44} At present, these are not suitable substitutes for the microchannel plate (MCP)/phosphor/camera combination for routine VMI use owing to cost and resolution limitations. Another 3D imaging approach has recently been developed by Li and co-workers.^{45,46} In this technique, the arrival time and position of the ion or electron are recorded by means of an inexpensive CMOS camera/high speed digitizer combination, with the time and position correlated in software relying on the natural pulse height distribution of the MCP. This has allowed for multielectron detection with a zero dead time and time- and position-correlation of twenty events or more in a single frame. This method thus shows some promise of being adapted for conventional VMI experiments.

In general, slice imaging relies on a fast switch to pulse the detector at the required arrival time. Given the capacitance of microchannel plate (MCP) detectors, a gate width of around 50–100 ns is readily achieved using widely available fast high-voltage switches. With some effort shorter gate times may be obtained, particularly for smaller detectors with lower capacitance, but 70 ns nominal gate width is routine in our work. This finite gate width is thus an implied limit to how effectively sliced an ion cloud can be, which is also directly related to the size of the ion cloud and its temporal spread on arrival at the detector.²¹ Slice imaging thus always delivers an approximation to the infinitesimally thin slice of the ion (or

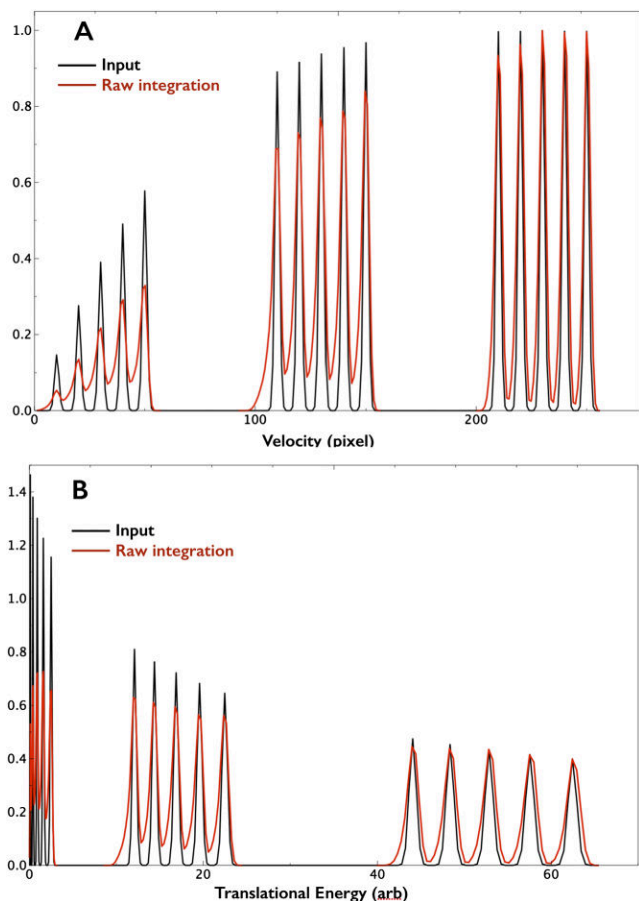


FIG. 1. Synthetic sliced data (20% sliced of 512 pixels). The black line corresponds to the input distribution and the red line is the raw integration of the sliced image. (a) Velocity distribution; (b) corresponding translational energy distribution (proportional to pixel^2 after applying the appropriate Jacobian).

electron) cloud. This finite slice width means that the recorded image deviates from the 3D distribution in subtle ways that can impact the resolution and the inferred velocity distribution of the image. This effect is illustrated in Fig. 1 using a synthetic image. Typically, the effective slice width that can be achieved is on the order of 10%–30% of the full temporal spread of the ion cloud. This is enough to recover the bulk of the velocity distribution accurately: as shown in Fig. 1, errors appearing at 10% in velocity are only 1% in translational energy, which is usually the distribution of interest. Nevertheless, at smaller radii the effective slice width becomes a much larger fraction of the recoil and eventually at very low radii it becomes impossible to slice effectively. The result is both compromised resolution and inaccurate weighting of the slow fragments, as seen in Fig. 1.

Hall and co-workers⁴⁷ developed a method to correct slice imaging data for imperfect slicing, and it has been applied to CO produced from ketene photodissociation, and to imaging of formaldehyde photodissociation.^{48–50} However, this approach is only suitable for data with isotropic angular distributions. In order to correct for the finite slice width for slice imaging data, and to apply to sliced or unsliced data without enforcing cylindrical symmetry, we have developed a new algorithm to correct for these out-of-plane contributions. We refer to this program as the “FINite sliced Analysis method” (FinA). We have developed two principal approaches: one uses a

spherical polar basis for the analysis and reconstruction, while the other employs a Cartesian basis. In this paper, we describe the polar treatment, while the Cartesian approach will be presented in a future publication. These have been developed as MATLAB routines using its graphical user interface (GUI), and the package will be made available to interested readers. In general, the method neither requires nor enforces full cylindrical symmetry. As we will show, it can also be used to determine the effective slice width, which is often difficult to measure directly. Furthermore, using synthetic data we evaluate and discuss the impact of various levels of slicing and added noise on the inferred distributions.

We present a number of illustrative studies to showcase the FinA method using synthetic images produced via a Monte Carlo based sampling method, as well as experimental images that result from the photodissociation of OCS at 230 nm and 217 nm recorded at a number of different slicing conditions. We also compare the reconstruction of sliced and unsliced data using FinA with that of unsliced images reconstructed using the popular BASEX method.¹² OCS was chosen as a well-studied system to illustrate the FinA method as the two co-fragments, S(¹D) and CO, can be readily produced and ionized using a single laser beam and give useful structured distributions.^{51–55}

II. COMPUTATIONAL METHOD

The chief purpose of the FinA method is to correct a time-sliced ion image in a way that accounts for the out-of-plane contribution arising from the finite slice width. We find that its performance on unsliced data is comparable to BASEX as well. To introduce the method, first let us consider an ideal situation of a simple unsliced isotropic sphere. In this situation, the projection of the sphere onto a 2-D plane results in a sharp, high intensity ring at a distance i from the center (where i is the radius of the sphere) with gradually decreasing intensity from the ring towards the center of the image. Knowing that the sliced image in this situation should consist solely of a single sharp ring, we assert that if the unwanted signal for $r < i$ can be modeled and removed from the projected image, and then the central slice distribution is what remains. We develop basis functions that model the unwanted out-of-plane elements recorded in the projection to permit a simple method of sequential subtraction to remove them. We will first operate in a spherical coordinate system (Fig. 2) in a *detector frame* where the Z axis is the time-of-flight direction, Θ describes the out-of-plane components with $\Theta = 90^\circ$ parallel to the detection plane, and Φ the azimuthal angle ranging from 0° to 360° . The angle $\Phi = 0$ corresponds to the electric field direction of the photodissociation laser or the relative velocity vector if we have a cylindrically symmetric distribution. We note that in this picture our azimuthal angle Φ relates to the polar angle θ commonly used to describe the product angular distributions, albeit with a range of 0° – 360° rather than 0° – 180° . In this case, the polar axis z to which θ is referred corresponds to the X axis in the detector frame (see Fig. 2). The initial assumption we employ is that the out-of-plane contribution for a given azimuthal angle Φ depends only on that angle at a given velocity and is independent of Θ . This is only

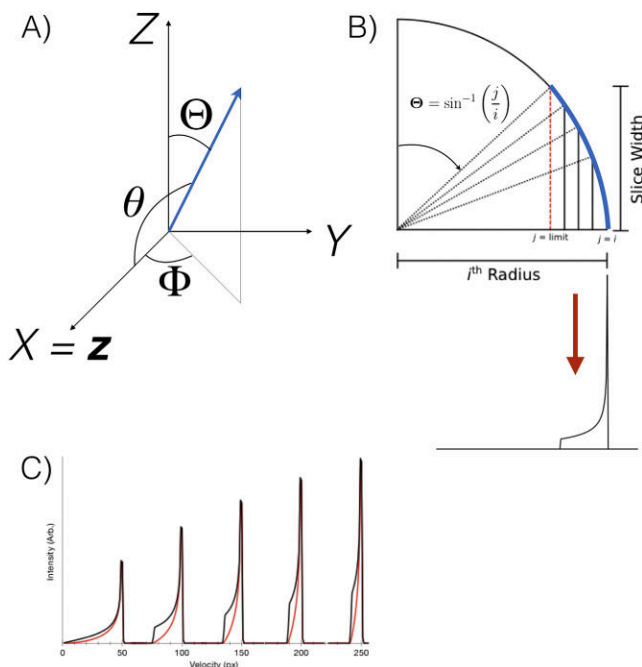


FIG. 2. (a) Coordinate systems employed and relation of variables. In our initial detector frame, the Z axis is the imaging time of flight direction and X-Y plane is the detection plane. For shaping the basis functions, we need the conventional geometry with the z axis, the polar axis in the plane. (b) Relation of the spherical polar angle and slice width used in establishing the basis functions. (c) A series of basis functions for different radii employing rectangular (black) or triangular (red) gates.

true for isotropic scattering but it allows us to develop “universal” zero-order basis functions describing the contribution of a sliced segment of a sphere at a given radius to all smaller radii. Although detection of the 2D projected image is done using a Cartesian coordinate system (usually a CMOS or CCD camera), we first convert from this to a polar grid by interpolation. As mentioned above, this is always done in any case to obtain the velocity distribution; here we simply do it prior to reconstructing the image.

In Fig. 2(b) we illustrate the nature of these basis functions. We can simply model this as the total area of the sphere that falls within a single bin (in δr and $\delta\Phi$) in the 2D projection plane. The construction of the basis function, as can be seen in Fig. 2(b), is given by

$$f^i(j) = j \left(\sin^{-1} \left(\frac{j-1}{i} \right) - \sin^{-1} \left(\frac{j}{i} \right) \right), \quad (1)$$

where i is the radius of the sphere and j is the radial bin for its projection onto the plane. The difference in the inverse sine terms represents the portion of the line integral along the sphere that falls into the bin, while j gives the appropriate weighting by the radius of the segment at that value of Θ . As shown in Fig. 2, for a sliced image there is a limiting radius within which there is no further contribution to the basis function for a particular slice width and sphere radius i . This limit L is given by $L = \sqrt{(i^2 - w^2)}$ when the slice width w (in pixels) is smaller than the total radius of the sphere (i.e., $i > w$), or $L = 0$ when this condition is not met. Figure 2(b) graphically depicts how all these variables are related. Each basis function is thus a vector labeled by the radius i . Figure 2(c) shows these

zero-order basis vectors at various radii for rectangular and triangular gates (*vide infra*).

At this stage, we can describe how the method will operate. After transforming the recorded image projection into a spherical polar coordinate system (with $\Theta = 90^\circ$), each angular bin (Φ) is treated independently in the following manner. Starting from the outermost radial bin and moving inwards, the first signal pixel is located at a radius i . Its intensity is divided by the intensity of the first pixel in the corresponding basis vector (appropriate for that radius and slice width). The resulting value is recorded in the corresponding bin in the output image. The basis vector is then weighted by the signal pixel intensity and subtracted from the original image for that angular bin. This in effect removes all out-of-plane contributions expected from that signal pixel. This process is then repeated on each successive nonzero pixel inwards towards the center and then repeated for each successive angular bin. This process is laid out as a flow chart in Figure 3.

For sparse images of course there may not be a signal that smoothly represents the out-of-plane contributions. Subtracting the basis vector may thus give negative values at some points in the image as it is being processed. Negative values in the subsequent image are similarly treated, with the basis functions now added rather than subtracted. In the end, as expected, these negative values generally disappear after integration to obtain the angular and velocity distributions.

So far we have described the basis functions appropriate for an isotropic angular distribution. We find that these “universal” spherical basis functions often give reasonably accurate reconstructions even for anisotropic images; however, for a rigorous approach we must account for the angular anisotropy in setting the shape of the basis functions. To do this requires some foreknowledge of the angular distribution

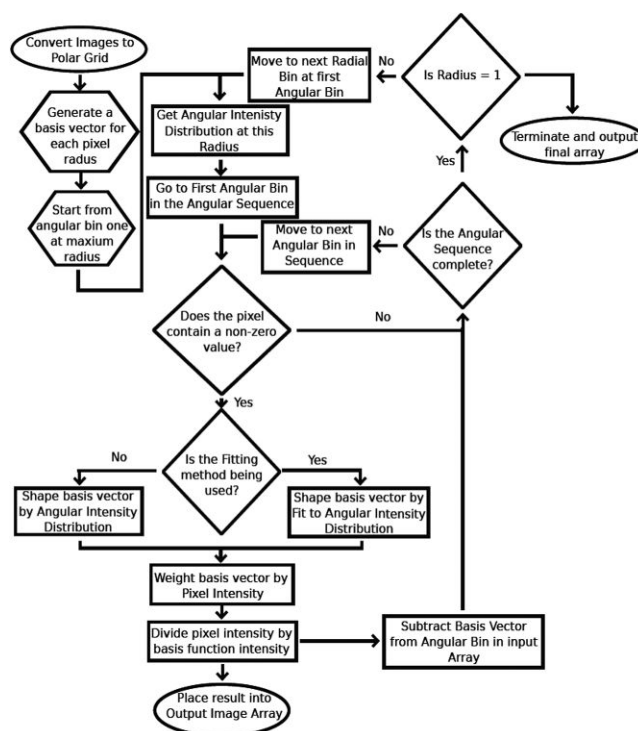


FIG. 3. Flow chart of the FinA approach.

at each radius because the out-of-plane contribution represents different polar angles θ (with respect to the z axis in the plane) that project onto the detection plane at a given Φ . We develop this in two fundamentally different ways depending on the problem: Either we simply use the intensity in the original image to determine the weighting of the out-of-plane contribution to the basis function at a particular value of Θ and Φ (the “General Polar” method), or we fit the entire angular distribution at each radius to an expected model, then use the fitted angular information to shape the basis functions prior to subtraction (termed the “Fitted Polar” method). The first approach is a very general method appropriate for any case, including arbitrary angular distributions as may be encountered in scattering experiments. It works because the input image, even unsliced, already represents a reasonable approximation to the sliced 3D distribution; indeed this is the foundation of the Vrakking iterative method. Here we simply use the original image to shape the basis functions for subtraction, and there is no need for further iteration. The fitting approach is suitable for photodissociation or photoelectron imaging when the angular distributions may be fitted accurately with a few parameters. It gives excellent performance for the velocity distributions when it is applicable.

In either of these approaches, to determine the correct out-of-plane weighting we first need to know the actual polar angle θ represented by a given (Θ, Φ) combination in determining the out-of-plane contribution. This angle is simply given by $\theta = \cos^{-1}(\sin \Theta \cos \Phi)$. For a given image pixel, as the out-of-plane contributions to the basis function are being evaluated, the angle θ is determined, and then the appropriate weighting is found either using the intensity in the original image at that value of (i, θ) or using the fitted angular distribution. In either case the basis functions must be determined separately for each point in the image, but as we do this either using the fitted angular distribution or simply by means of a look-up using the value in the original image, it is not computationally demanding.

In detail we proceed as follows after first interpolating the image onto a polar grid at a user-selectable angular resolution: For the general polar approach, at every point in the polar image we select the appropriate spherical basis function $f^i(j)$. Each element of the basis function is then weighted to shape it according to the underlying angular distribution,

$$f^i(j, \Phi) = s(i, j, \Phi) f^i(j), \quad (2)$$

where the weighting is given by the intensity in the original image, $I(i\Phi)$, at the appropriate location

$$s(i, j, \Phi) = I\left(i, \cos^{-1}\left(\frac{j}{i} \cos \Phi\right)\right). \quad (3)$$

To achieve the fitted polar method in practice, we first fit the full angular distribution in the i th radial bin prior to any subtraction. This is performed using an expansion in Legendre polynomials up to fourth order, although often the well-known second order expression serves for most photodissociation or photoelectron imaging experiments,

$$I(i, \theta) = \beta_0(i) (1 + \beta_2(i) P_2(\cos \theta)). \quad (4)$$

In the experiments described below, this accounts for the angular distribution resulting from a single photon dissociation

event. In this equation, the parameter β_0 is the intensity dependent parameter, β_2 is the normalized anisotropy parameter, and P_2 is the second order Legendre polynomial. From this point, the weighting $s(i, j, \Phi)$ is then given by the following equation:

$$s(i, j, \Phi) = \beta_0(i) \left(1 + \beta_2(i) \left(3 \left(\frac{j}{i} \cos \Phi \right)^2 - 1 \right) / 2 \right). \quad (5)$$

Then, as above, by scaling each point in the basis function $f^i(j)$ by $s(i, j, \Phi)$, we produce a new shaped and weighted basis function. This vector is subtracted from that angular bin just as described previously. This process is repeated for all angular bins at radius i , then $i - 1$, and so on to the center of the image. In either case the final distributions are directly obtained from the polar output image. It may be then projected onto a Cartesian grid for comparison to the original image if desired.

So far we have assumed that the slice gate applied to the detector is rectangular so that the cutoff is sharply defined at a particular radius. However, it is readily seen experimentally, as we show below, that this is not the case. Time-slicing is generally performed by pulsing the MCP detector using a high-voltage switch during the arrival of the ion sphere. Given the capacitance of a typical MCP detector, one would not expect to be able to achieve nanosecond rise times as would be necessary to obtain a rectangular gate on a time-of-flight peak that is 100-200 ns wide. In practice, for short pulses we find the detector gain profiles to be roughly triangular, which is advantageous for fast slicing in any case. This effect must be considered in modeling the distributions to generate the correct basis functions, regardless of which method above is used. For pulses of intermediate duration, the correct description would be achieved using a trapezoidal gate. In practice however, as will be shown below using experimental results, the triangular gate approximation is found to be a good description of the gain profile for a pulsed MCP. We thus limit our approach to consider either triangular or rectangular gating as an option. The difference between the rectangular or triangular gating is highlighted in Fig. 2(c).

In testing these reconstruction methods, in addition to experimental data, we use synthetic images produced using a Monte Carlo code that employs uniform sampling of a sphere using the Archimedes theorem. An input velocity distribution is assumed and sampled, and an assumed velocity-dependent angular distribution is then used to weight the intensities of the recorded events. The slice width is also readily adjusted, and rectangular gating is applied for the synthetic data. As seen below, we employ demanding sharply structured distributions and rapidly changing angular distributions to investigate the performance of the FinA method. We use images for which the velocity distributions grow linearly so that the corresponding images may be viewed clearly. Synthetic images are produced on a Cartesian grid matching typical experimental data.

Computational times were evaluated on a Macbook Pro with a 2.5 GHz Intel Core i7 processor with 16 GB DDR 3 RAM running MacOS 10.12.3 and MATLAB version R2015a. Using a 510×510 pixel image and one angular bin per degree, the basic FinA method required 1.6 s to reconstruct the image, while the fitted version required 35 s. Increasing the angular bin size increases the computational time

proportionately for the basic method but has only a modest effect on the time for the fitted method.

III. EXPERIMENTAL

In order to investigate and validate the FinA method with experimental DC sliced images, images were obtained for both fragments of the UV photodissociation of OCS. The experiments were performed using our standard DC slice imaging setup which has been described in more detail elsewhere.³ A seeded molecular beam of OCS (<0.1%) in He at a backing pressure of 1 bar was produced using a pulsed solenoid valve operating at 10 Hz with an opening time of 400 μ s. The molecular beam was skimmed using a 1 mm skimmer prior to entering the interaction region between the repeller and first electrode of an ion imaging apparatus. The molecular beam was then intersected at a right angle by a single laser beam. For recording the CO fragments, photolysis of OCS and 2 + 1 REMPI of the CO fragment via the *B* state was performed using a tunable laser beam around 230 nm at $\sim$ 0.35 mJ/pulse. This beam was generated by mixing 653 nm produced by an Nd:YAG-pumped dye laser with the third harmonic output of the same seeded Nd:YAG laser in a BBO crystal. The laser beam passed through the Berek compensator to rotate the polarization to be parallel to the detector and it was focused into the interaction region

using a 30 cm fused silica lens. The laser was scanned over the Doppler width $J = 47$ transition of ground state CO. For recording images of the S(¹D) fragment, photolysis of OCS and 1 + 1 REMPI of the S atom was performed at 217 nm at a power of 0.45 mJ/pulse. This beam was produced by doubling the 651 nm output from a dye laser (Cobra-Stretch, Sirah) in BBO and then mixing the UV with the dye fundamental in the second BBO crystal. Once again, the beam was passed through the Berek compensator prior to being focused into the chamber through a 30 cm fused silica lens.

The CO⁺ or S⁺ ions were extracted from the interaction region using an array of ion optics containing a repeller electrode and four lens electrodes, with voltages of $V_R = 646$, $V_{L1} = 611$, $V_{L2} = 558$, and $V_{L3} = 500$ V (L_4 was grounded), and accelerated towards the imaging detector comprising a 75 mm dual microchannel plate coupled to a P-47 phosphor screen (BOS-75-OPT01, Beam Imaging Solutions). The detector was gated by switching the potential on the back plate from 1.2 kV to 2.1 kV using a high voltage (HV) pulser (DEI, PVX-4150 pulse generator). CO images were recorded in this manner with a number of different MCP detector gates: nominally 75 ns, 120 ns, 160 ns, 200 ns, 250, 300, and 350 ns to study the effect of different effective slice widths. For the sulfur fragment, only the finely sliced (75 ns) and unsliced (350 ns) gate conditions were recorded. In addition to the analysis of the OCS data,

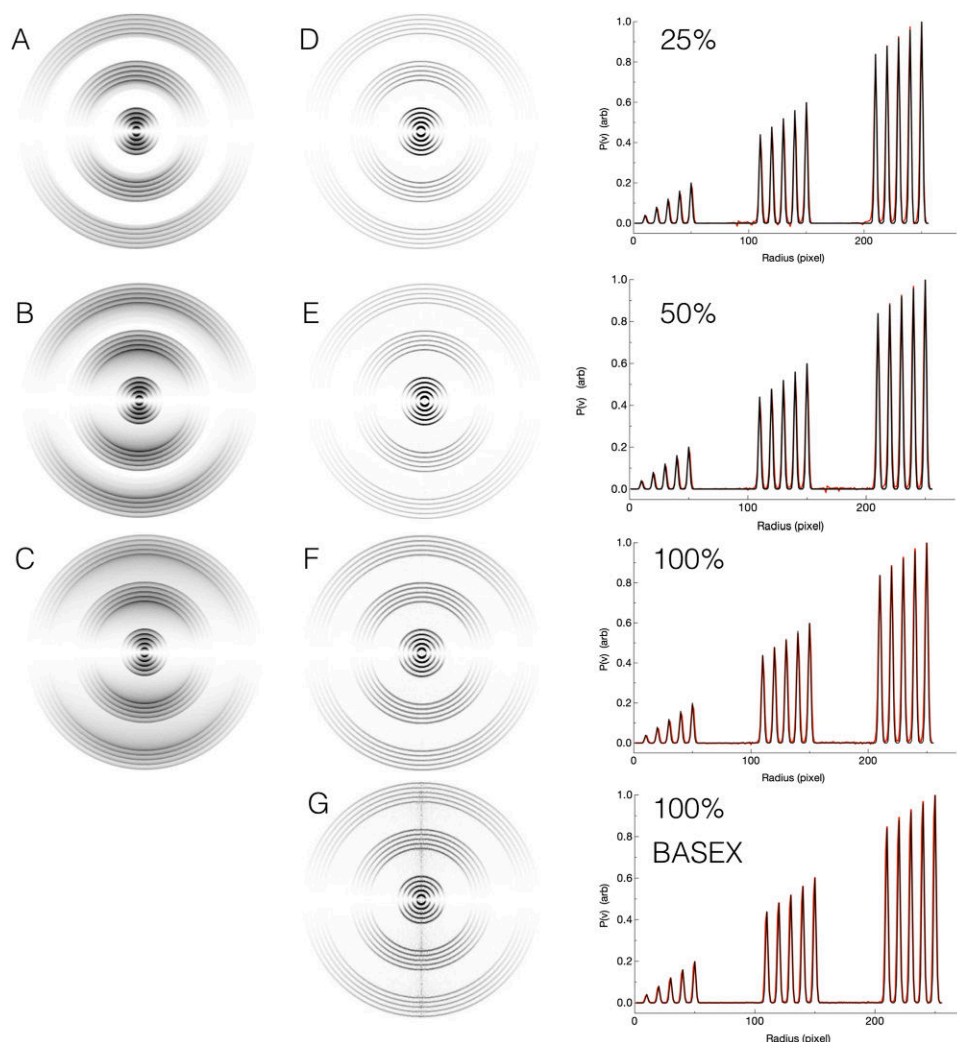


FIG. 4. Synthetic images and spectra produced at a variety of slice widths. Panels (a)–(c) show the raw images with slice widths of 25%, 50%, and 100%, respectively. Their corresponding FinA reconstructions are shown in panels (d)–(f). Panel (g) shows the BASEX reconstruction of image (c). The integrated spectra (red) for each of the reconstructions are shown to the right of the images. The input distribution is also overlaid in black in each of the spectra.

we also include scattering of NO ($V = 0$, $J = 14.5$) off argon. The apparatus for this measurement has been reported previously;²⁰ however, some additional details are listed here. NO seeded at 10% in Ar (at 50 psi) was introduced into the scattering region as a skimmed molecular beam. This collided off a skimmed molecular beam of pure Ar (at 80 psi). The distribution was probed by 1 + 1 REMPI on the R1 (14.5) transition of the A state at $44\,300.94\text{ cm}^{-1}$ and acquired for 3000 shots using a Nd:YAG pumped dye laser system. The ions produced were focused onto a 75 mm MCP/(P47) phosphor detector using a set of electrostatic lens with electric fields of 750 V and 700 V across the repeller and extraction lens.

IV. RESULTS AND DISCUSSION

In order to evaluate the FinA method, we have explored a number of different applications with both synthetic and experimental data. These include the following: (i) performance of the method when reconstructing data with a variety of different slice widths; (ii) the effectiveness of the reconstruction in recovering the input angular distribution parameters; (iii) how signal level and background noise influence the outcome of the reconstruction; (iv) the benefits of slice imaging coupled with the FinA method over unsliced approaches; and finally (v) how the method fairs with images lacking clear cylindrical symmetry. When making comparisons to unsliced data, we have selected the BASEX method as the standard and one of the most commonly used and high-performance reconstruction methods. Therefore, the comparison will always be made between BASEX or FinA transforming an unsliced image and FinA reconstructing a sliced image.

A. Slice width

One of the most important characteristics of a slice imaging experiment is the effectiveness of its slicing. This can be characterized as a “slice width,” i.e., the percentage of the sphere that falls within the recorded volume. Ideally, one would aim to have as small a slice width as possible to approach a direct measurement of the 3D distribution, affording increased resolution, improved signal to noise levels, and overall quality of the spectra. We also note that for ease of comparison here the slice width is quoted as the percentage of the largest radius sphere considered in the reconstruction. The slice width is also used to define the lower length limit of each basis vector.

The FinA method has been designed to reconstruct images of arbitrary slice width, and we will show that it may be used to determine the effective slice width for experimental images where this is not known. In order to investigate this first we consider a number of synthetic images shown in Figure 4. In this case, through the Monte Carlo simulation we are able to specify the exact slice width of the image. Shown in the left hand side of the image (Figs. 4(a)–4(c)) is a series of images with increasing slice widths. These are 25%, 50%, and 100% of 512 pixels (the size of the image). The images shown in Figs. 4(d)–4(f) are the corresponding reconstructions of the images (a)–(c), respectively. By visual inspection, clearly the images have become sharper (especially apparent in the unsliced image) and the angular distribution of intensity around the rings ($\beta_2 = 2$) is

maintained. The right hand side of Figure 4 shows the integrated spectra of the reconstructions in red. This is overlain with the input distribution (shown in black) used to generate the images. It is clear that in all cases the input distribution is recovered nicely, in terms of both the overall resolution of the peaks and the overall intensity distribution. In addition to the FinA reconstruction, the reconstruction of the unsliced image using the BASEX algorithm is also shown in Fig. 4(g). From the corresponding spectrum, it is clear that the image has been correctly reconstructed and the same intensity distribution and resolution are recovered. There is some noise along the centerline in the BASEX image that arises from the fact that the data there are less well determined, but the distributions are accurately recovered in all cases.

We note that use of the correct slice width is important to recover the spectrum accurately. This is highlighted in Figure 5, in which are shown spectra from FinA reconstruction of a synthetic image produced as a single sharp ring with a 35% slice. In the figure, a series of integrated spectra resulting from different, and incorrect, assumed slice widths are shown at $\pm 5\%$ and $\pm 2\%$ of the correct value. It is evident that the spectra are incorrectly recovered when the wrong slice width is used. This is manifest as negative peaks and other artifacts introduced into the spectrum. Typically, when using a slice width larger than that required, the image is effectively over-subtracted due to the extended basis vector. On the other hand, when using one too small the function is not sufficiently extended to account for all the projected elements

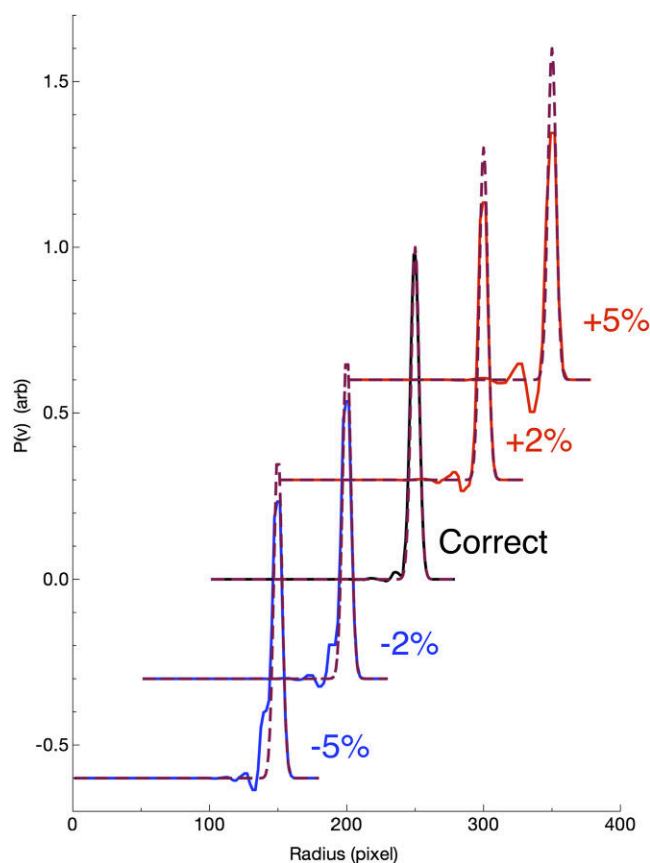


FIG. 5. Integrated spectra of a single ring with a radius of 200 pixels (red). Each reconstruction has been performed with a slice width of $35\% \pm 2\%$ or 5% . In each case, the input distribution is (dashed) superimposed on the result.

arising from the pixel and the image is under-subtracted and “ringing” occurs.

In practice, we find that even without knowledge of the slice width from the experiment, determining the slice width is possible after the fact. To demonstrate this, Figure 6 shows a number of CO images recorded from the photodissociation of OCS recorded with a range of gate widths. As these data were recorded using a pulsed MCP detector, a triangular weighting was applied to the basis function as discussed in Sec. II of the paper. Images (a)–(c) in the figure display the CO images recorded with nominal applied gate widths of 75, 200, and 350 ns. As evident from the raw images, as the gate width increases the contribution of out-of-plane signal increases until the image is fully unsliced (Fig. 6(c)). The image consists of a large amplitude isotropic ring with a small shoulder at larger radii. Images (d)–(f) in the same figure show the corresponding FinA reconstructed images. It should be noted that due to fluctuations within the image itself, there are a number of negative counts within the reconstructed image that are difficult to show effectively. Therefore, although the central region of the image in (e) and (f) seems to contain a significant amount of intensity, when integrated over all angles this is averaged out and largely disappears. In the FinA reconstructions, we also note that any noise within the image accumulated during the reconstruction accumulates towards the center of the image. Although this may seem disadvantageous, due to the $r \sin(\theta)$

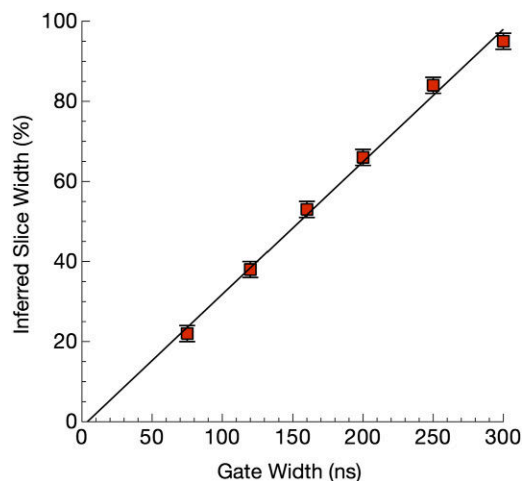


FIG. 7. Inferred slice width of CO data plotted as a function of detector gate width.

weighting of the intensity in producing the integrated spectra (shown in the right hand panel of the figure) this noise does not contribute strongly to the spectra.

In addition to the FinA reconstruction, we have also shown the BASEX reconstruction of the unsliced data. From all the spectra presented, it is clear that the reconstruction successfully recovered a single sharp, essentially isotropic ring with a FWHM speed distribution of $53 \pm 2 \text{ ms}^{-1}$ (blurring

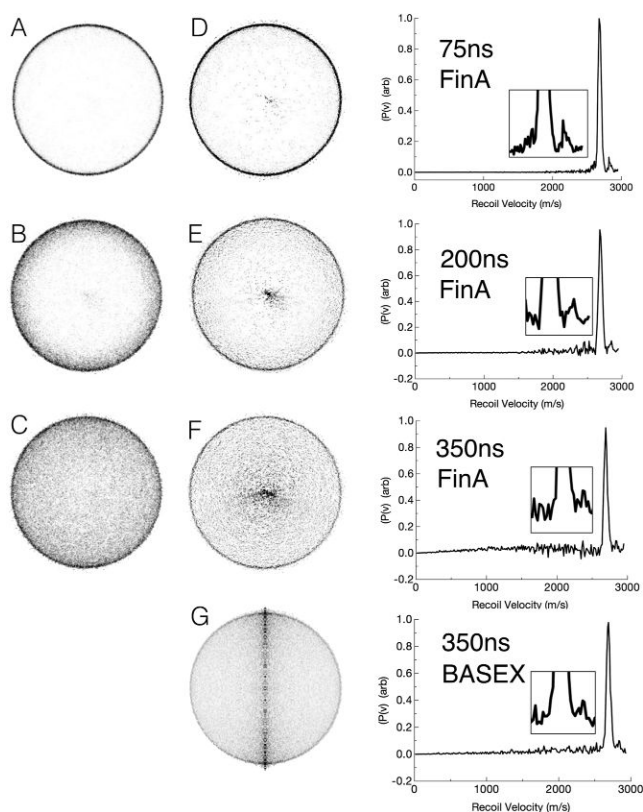


FIG. 6. $\text{CO}(X^1\Sigma^+)$ fragment images resulting from the photodissociation of OCS at 230 nm are shown as a function of the detector gate width. The raw images are shown in panels (a)–(c) and their corresponding reconstructions in panels (d)–(f). The BASEX reconstruction of the unsliced image is shown in panel (g). The resulting integrated spectra are shown to the right of the reconstructed image. An inset in each panel highlights the small feature at a higher recoil velocity of the main peak.

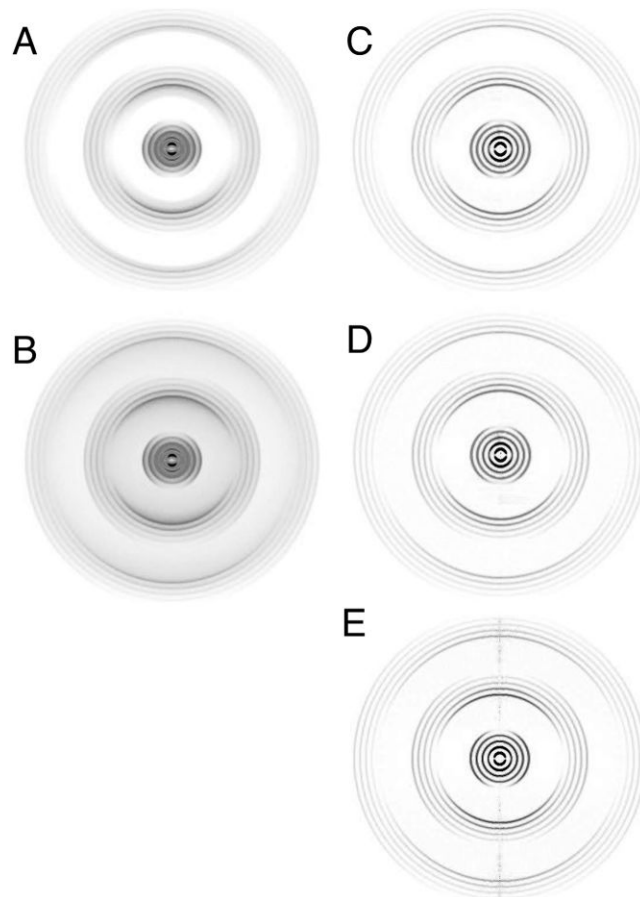


FIG. 8. Synthetic images with rapidly changing angular distributions. Panels (a) and (b) show the raw sliced and unsliced images. Panels (c) and (d) display the corresponding FinA reconstructed images. Panel (e) shows the BASEX reconstruction of the unsliced image.

slightly to 58 m/s in the unsliced image (FinA and BASEX)). This is just slightly larger than that expected for the cation recoil in the REMPI probe, ~ 35 m/s. A notable aspect of these data is not just that the reconstruction is successful but also the clarity and sharpness of the reconstructed slice data over the unsliced case. Keeping in mind that all images were recorded using the same conditions (aside from the different gate width across the MCP detector) and over the same amount of time, it is clear that the sliced image reconstruction S/N is significantly better than the unsliced reconstructions (FinA or BASEX), is somewhat sharper (as evident from the narrower speed distribution), and the baseline is much smoother. Furthermore, at a slightly higher recoil velocity than the main peak, there is a smaller feature that is only completely resolved in the sliced image, shown magnified as an inset in each spectrum. This corresponds to CO recoiling from ^{34}S as opposed to the more abundant ^{32}S isotope responsible for the main peak. Based on the integrated intensity of the feature, the faster peak constitutes 4.24% of the total CO signal, which is well within the range 3.98%–4.73% for the natural abundance of this isotope.⁵⁶ We should add, however, that in practice these CO images would likely require no reconstruction at all, since they have no low-velocity components. However, they provide a very useful experimental test case.

In addition to reconstructing the images accurately, we are able to determine the effective slice width of these images as a function of gate width by applying the method and varying

the assumed width. Figure 7 shows the results for the images shown in Figure 7 (75, 200, and 350 ns) along with slice widths inferred in analyzing additional images with applied gate widths of 120, 160, 250, and 300 ns. It is clear from Figure 7 that the distribution follows a linear trend as a function of increasing gate width. Interestingly, the x -axis intercept (i.e., when the slice width is 0%) is found at 4 ns, which is very close to the temporal pulse width of our probe laser.

B. Angular distributions

An important challenge for the method is the ability to accurately reconstruct images with angular distributions that vary and even vary rapidly. Therefore, a rigorous test was performed to examine how FinA handles images with rapidly changing angular distributions.

Figure 8 displays a particularly complex set of images with sharp peaks and regularly changing angular distribution. Figure 8 panels (a) and (b) show the raw synthetic images with slice widths of 25% (sliced) and 100% (unsliced) at 512 pixels with their FinA reconstructions displayed in (c) and (d), respectively, along with the BASEX counterpart displayed in panel (e). It is clear in all cases that the images (at least visually) have been correctly reconstructed and the angular distributions remained intact. To test this rigorously, however, the images were subsequently fitted using a 2nd order Legendre polynomial expansion at each radius and the fits to this are plotted in Figure 9. The solid lines in panels (a) and (b) represent

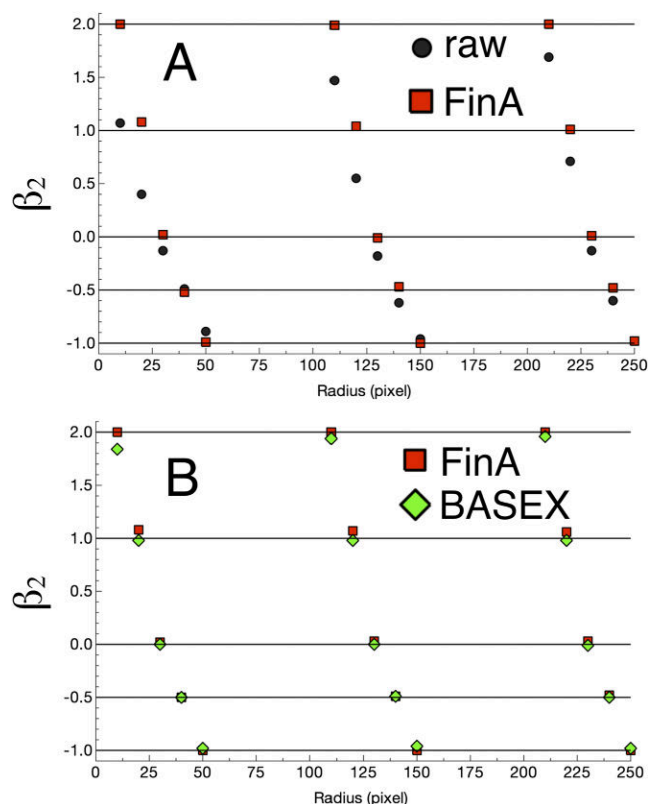


FIG. 9. Fitted β_2 anisotropy parameters of the images shown in Figure 8 plotted as a function of velocity. The input parameter is shown as a solid line in both panels. Panel (a) shows the raw (black) and FinA (red) fitted β_2 parameters as a function of radii for the sliced image. Panel (b) shows the FinA (red) and BASEX (green) fitted β_2 parameters as a function of radii for the unsliced image.

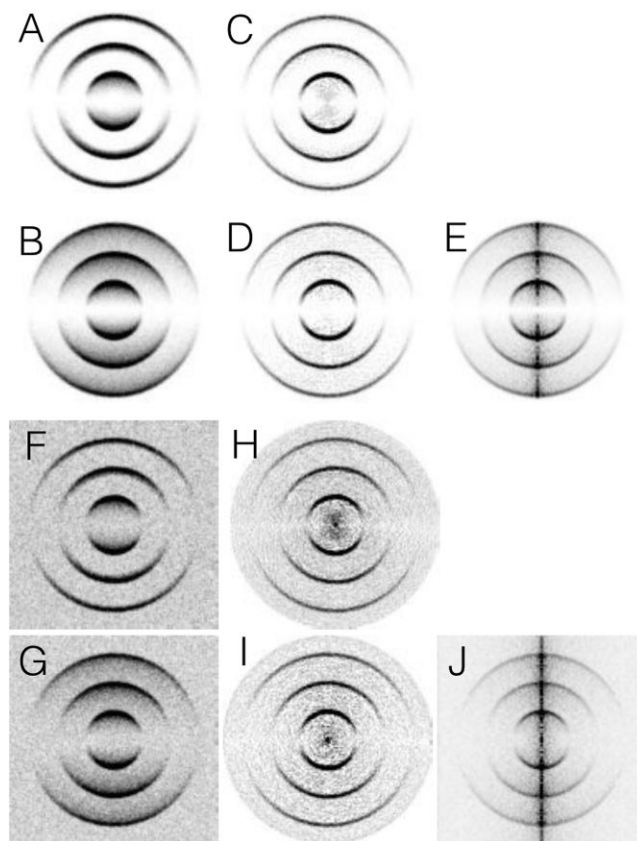


FIG. 10. Synthetic images produced with 10^6 input events. Images (a) and (b) show the input sliced (25%) and unsliced images. Their corresponding FinA reconstructions are shown in panels (c) and (d) alongside the BASEX reconstruction in panel (e). Panels (f)–(j) show the images and their corresponding reconstructions after background noise has been added.

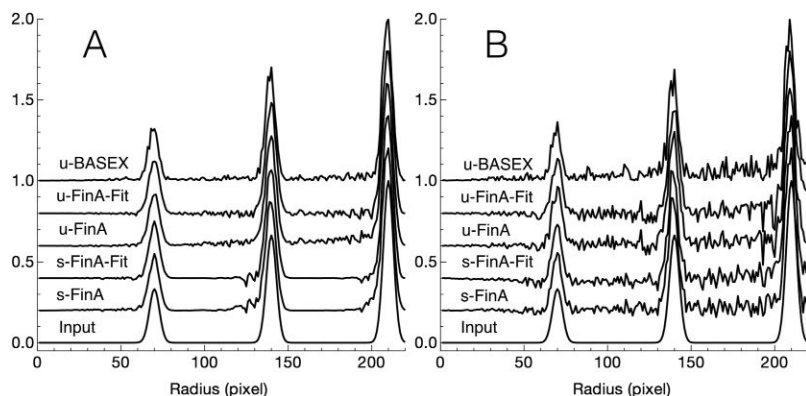


FIG. 11. Spectra showing the reconstruction of the low count data displayed in Figure 10. Panel (a) shows the reconstruction of the images with no noise. Panel (b) shows the reconstruction of the images with noise. Methods noted with an “s” are reconstructions of sliced images whilst “u” represents unsliced images.

the input angular distributions, which go in order of $\beta_2 = 2, 1, 0, -0.5$, and -1 in each set of rings. Panel (a) shows the results from the sliced image reconstruction, and the unsliced is shown in panel (b). Panel (a) shows that the simple reconstruction of the raw sliced image is reasonably successful in capturing the input distribution within the outermost two peaks of each set. However, even at the slice width of 25%, due to the contribution from the outer peaks, the inner peaks’ angular distributions become somewhat swamped and distorted. It should be noted that this is a more demanding example than is likely to be encountered in an actual experiment. In any case, the FinA reconstructed image is very successful in recovering the angular distributions. As shown in the panels (a) and (b) (red squares) in both the sliced and unsliced cases, the FinA reconstruction is accurate to within 0.05 of the input β_2 parameter, which is also accurately recovered using the BASEX method. It is interesting that the FinA method shows the greatest error for slightly parallel distributions, while BASEX deviates more at the limiting values and for the low recoil velocities where the noise line becomes an issue. This demanding test shows how robust both the FinA and BASEX methods are in handling challenging synthetic data.

C. Signal level and noise

Another important aspect of imaging performance is the reconstruction of sliced or unsliced data in the presence of low counts and background noise. To investigate this, we turn once again to some synthetic data but in this case we have reduced the number of input event used in the Monte Carlo simulation from 10^{10} down to 10^6 . At this point, we expect to see that statistical noise which develops purely from insufficient events needed to adequately describe the true 3D distribution, much as would be encountered in typical experiments. In this case, we also examine the performance of the FinA-Fit version of the method. In this approach all signals at a given radius

TABLE I. Sum of the squared residuals of the spectra shown in Figure 11 from the input distribution.

	Sliced no noise	Sliced noise	Unsliced no noise	Unsliced noise
FinA	0.27	0.76	0.38	1.14
Fitted FinA	0.14	0.70	0.15	0.70
BASEX	...	...	0.47	1.47

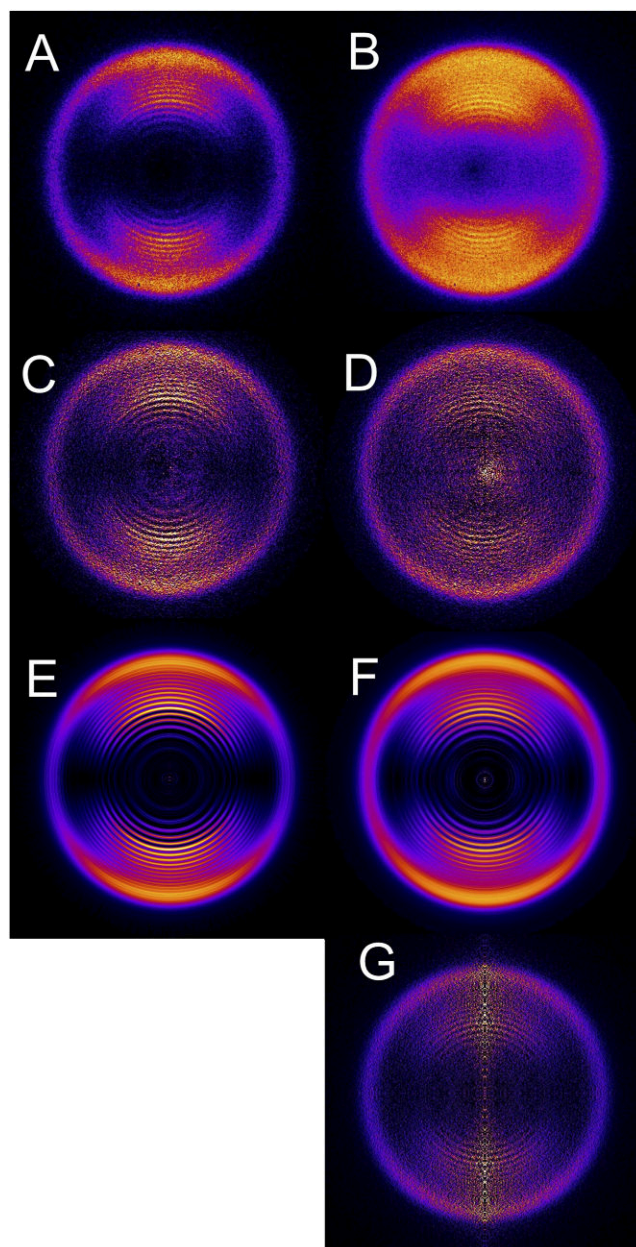


FIG. 12. S^1D fragment images resulting from the photodissociation of OCS at 217 nm. (a) and (b) are sliced and unsliced experimental images, respectively. (c) and (d) are FinA reconstructions of (a) and (b). (e) and (f) are FinA-Fit reconstructions of (a) and (b), and (g) is the BASEX reconstruction of (b).

are fitted, and then the fit is used to determine the subtraction of the basis vectors at each point on that radius, resulting in improved S/N for low count data. The images are shown in Figures 10(a)–10(e) with the resulting spectra shown in Figure 11(panel (a)). The images were generated using three Gaussians as the input distribution. The fit version gives excellent performance both with sliced and unsliced data as shown by the summed square of the residuals given in Table I. The basic FinA version gives somewhat better performance for the sliced image than for the unsliced case, and for unsliced images, FinA gives better performance than BASEX. This may be because the noise in the FinA method is scattered around zero, while in BASEX it is entirely nonnegative. However, we are reluctant to draw definitive conclusions about the performance of FinA relative to BASEX on unsliced data without more detailed investigation. The performance of FinA with sliced vs. unsliced data points to one of the real advantages of slice imaging, which is not well-captured using “perfect”

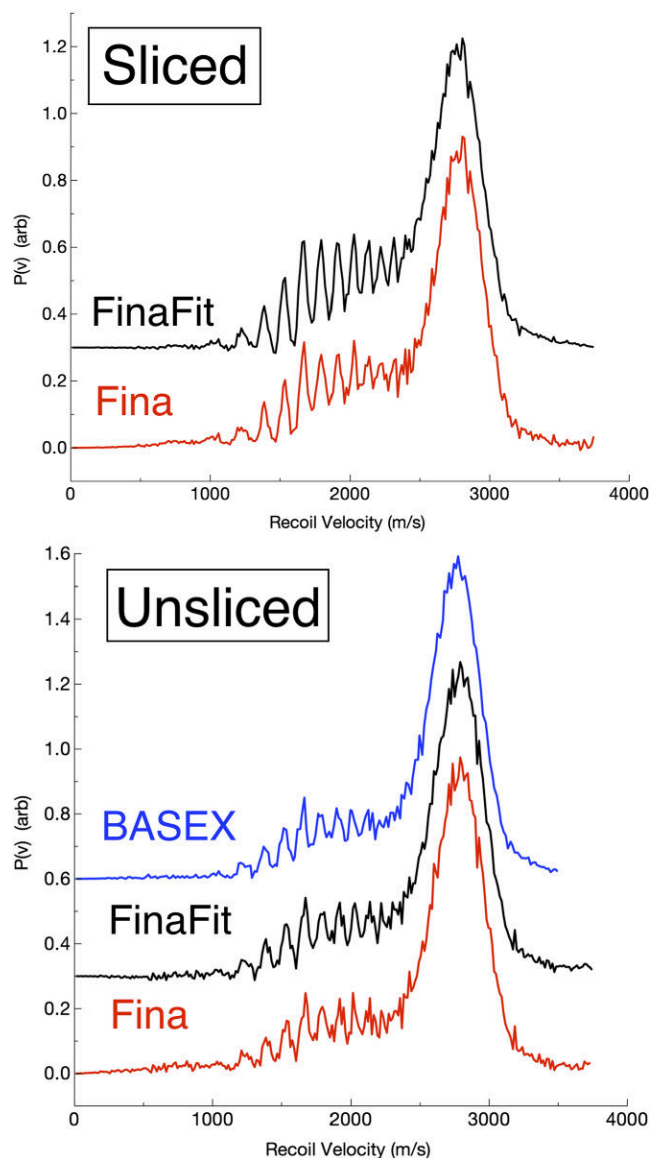


FIG. 13. Total relative velocity distributions of $S(^1D)$ from images in Fig. 12. The upper panel displays the results from the sliced image, whilst the lower displays the unsliced results. In each panel, the general FinA results are shown in red, the fitted-FinA results in black, and the BASEX reconstruction in blue.

synthetic images with very high counts. In sliced images, the data counts are concentrated in regions more strongly correlated with the underlying distributions, so the reconstructions are less prone to mapping the statistical noise into the distributions. The noise that appears in the gaps in the unsliced results, but is absent in the sliced data, shows this very clearly. This is also why the experimental data here show much better performance for the sliced data compared to the unsliced data.

We have also modified these low count images with a positive half-Gaussian distribution of noise. The distribution

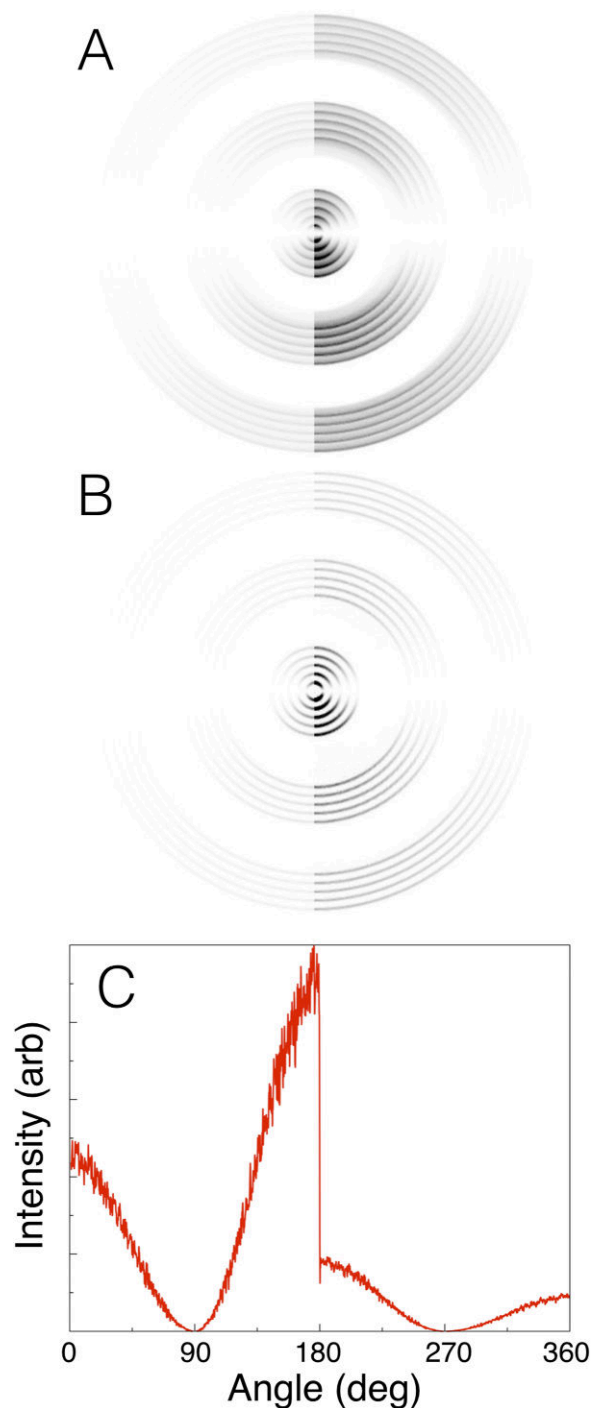


FIG. 14. Panel (a) shows a raw scaled synthetic image (see text for details). Panel (b) displays the reconstruction using the general FinA method. Panel (c) shows the angular distribution of intensities from the reconstructed image, integrated over 205–255 pixels.

places positive counts with a standard deviation of 3 counts across the entire image. This is approximately five times the average pixel intensity within the sliced image (25% of 512 pixels) and 2.5 times the average pixel intensity within the unsliced image and much more noise than would be encountered in a typical experiment. We kept the total noise the same in the sliced and unsliced cases to explore the situation in which the noise is a fixed function of acquisition time and independent of the gate width. The imaging results are also shown in Fig. 10 (panels (f)–(j)) with the recovered distributions also in Fig. 11 (panel (b)) and the residuals in Table I. In this case, the sliced result for the basic FinA is substantially better than the unsliced case (FinA or BASEX). Here the advantages of the fitting version of the program are clearest on the unsliced data, for which the summed residuals are identical to the sliced case.

D. Performance of DC slice imaging

Now that we have demonstrated the use of FinA method and its handling of a number of test cases, we turn to other examples to highlight the main advantage of the combination of FinA and DC slice imaging over its more traditional counterparts. For this we have recorded a sliced (35% of 550 pixels) and unsliced image of $S(^1D)$ recoiling from CO in the one-color photodissociation of OCS at 217 nm. Both of these images were recorded with the same parameters and conditions with the exception of the gate width applied to the MCP of 75 ns and 350 ns, respectively. The raw images and their corresponding FinA and BASEX reconstructions are shown in Figure 12. In addition to the raw images ((a) and (b)) and FinA reconstructions ((c) and (d)), we have also included a fitted FinA reconstruction. These fitted reconstructions are shown

in panels (e) and (f) (sliced and unsliced, respectively). In (e) and (f) we show only the result of the fit, much as is done by pBASEX. Finally we have also performed a BASEX reconstruction of the unsliced data, which is shown in Figure 12(g). The $S(^1D)$ image contains a bimodal distribution with a number sharp rotational features superimposed upon it. As discussed in previous work,^{51–55} these sharp features arise from a number of rotational levels produced in the CO co-fragment. In all cases, it is clear that both the angular distribution and the sharp features in the radial distribution are clearly resolved in all the reconstructions of both the sliced and unsliced data. However, the details of the reconstructions are shown in the spectra plotted in Figure 13. The upper panel depicts speed distributions obtained from the reconstructed sliced image using both the basic FinA (red) and fitted FinA methods (black). In each case, it is clear that the superimposed peaks are extremely well resolved with an obvious rotational progression. Clearly the fitted FinA method improves on the resolution, especially in the region between 2000 and 2500 ms^{-1} . The biggest contrast, however, is when comparing the sliced data to that of the unsliced data. It is clear that both FinA and BASEX give very similar results when treating the unsliced data. However, the unsliced spectra are clearly not as well resolved and are noisier than the sliced spectra. As both images were recorded under the same conditions and for the same amount of time, this clearly demonstrates the advantages of slice imaging. Similar enhanced performance for sliced imaging is seen in the insets to the CO spectra in Fig. 6 as discussed above.

E. Breaking cylindrical symmetry

The final point we wish to make about the FinA method is that it does not require complete cylindrical symmetry about

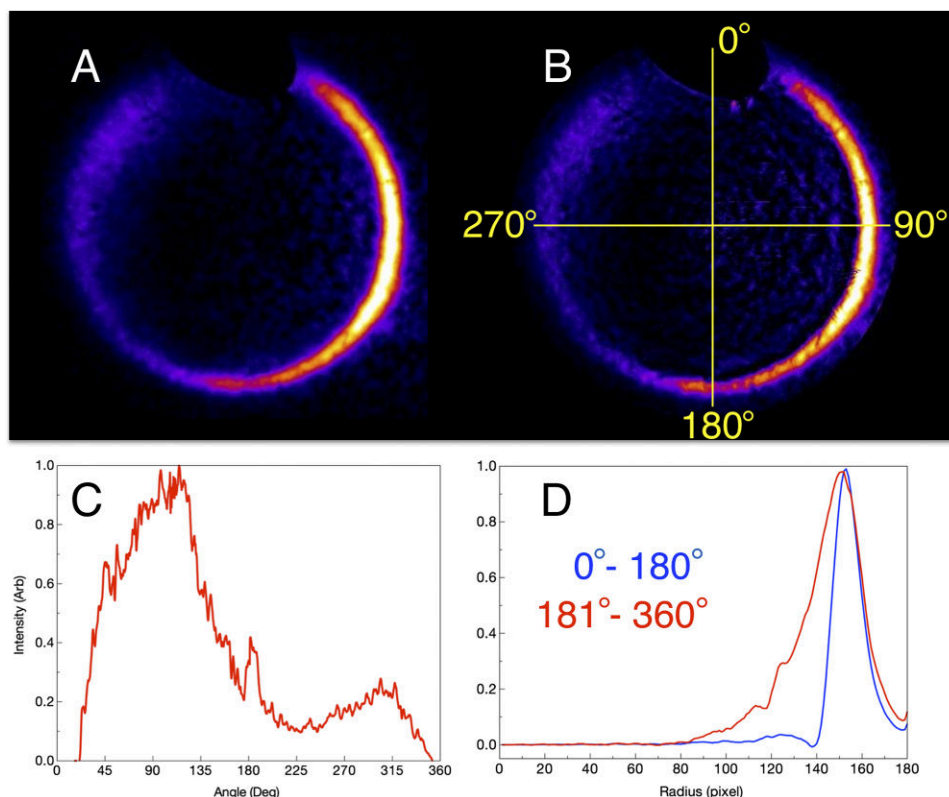


FIG. 15. Crossed beam scattering data of NO off Ar probed at $J=14.5$ in $v=0$. Panel (a) shows the raw scattering image. The beam spot has been masked post collection. The reconstruction using general FinA is displayed in panel (b). Panel (c) shows the differential cross section of these data integrated over the entire peak. A guide of the quadrants has been superimposed in panel (b). Panel (d) shows the angle-integrated spectrum of the 0°–180° side (blue) and 180°–360° side (red) of the image.

the initial 3D distribution, although each half of the image is assumed to exhibit local cylindrical symmetry in shaping the basis vectors. This can be simply shown using a modified synthetic image, based upon the sliced image shown in Figure 4(a). Each quadrant of the image has been linearly scaled independently to break the symmetry. This image is shown in the upper panel of Figure 14. For reference the upper left quadrant remains the same, the lower left has been scaled by a factor of two, the upper right by a factor of five, and finally the lower right quadrant has been scaled by a factor of 10. Clearly from the reconstruction it is easy to see that the image has been correctly reconstructed. In fact, as shown at the bottom of Figure 14, the angular distribution, summed over the outer 5 peaks, clearly shows that the angular distribution is as would be expected.

We provide a further demonstration of the treatment of asymmetric images with an experimental scattering image shown in Fig. 15(a). These are data for NO($v = 0$, $J = 14.5$) resulting from scattering a beam of NO off argon. The NO beam is heated via flash pyrolysis to produce NO($v = 1$) for scattering experiments to be described in a future publication.⁵⁷ The data have not been corrected for the density-to-flux transformation and because of this and the velocity spreads of the beams, there is a substantial asymmetry in the image. The image has been rotated so that the relative velocity vector is parallel to the z -axis. Panel (b) shows the FinA reconstructed image. From visual inspection it is clear that the reconstruction is successful and no anomalies or artifacts are introduced. We find that the experimental setup used to acquire the image has an effective slice width of 58% at 170 pixels. Panel (c) shows the angular distribution obtained from the reconstruction showing similar differential cross sections in each half but with significantly differing magnitudes. The integrated recoil velocity distribution is shown in Figure 15(d) for each half of the image separately. The sharper distribution is from the side that has a lower laboratory velocity, as a result of compression of the velocity spreads.⁵⁸

V. CONCLUSION

We have demonstrated our new method, FinA, for reconstructing partially sliced or unsliced imaging data valid for VMI, scattering, DC slicing or time-sliced imaging, and (although not explicitly discussed within this work) photoelectron imaging. We have shown that the method is successful at reconstructing a range of synthetic and experimental images with performance that equals, if not exceeds, typical results obtained through velocity map imaging coupled with standard methods. We show, with experimental data in particular, that the FinA reconstruction of sliced images offers clearly enhanced resolution and signal-to-noise when compared to unsliced data recorded under the same conditions.

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